

STRATEGIC RESEARCH DIRECTION REPORT

CogProfile-Net

Uncertainty-Aware Neural Cognitive Profiling via Riemannian EEG
Encoding with Psychometrically-Grounded Career Aptitude Mapping

Hardware: Emotiv EPOC+ (14ch, 128 Hz) **Subjects:** ~25 participants

Classes: 5 cognitive tasks **Baseline:** 73.34% (LightGBM)

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Table of Contents

1. Where We Stand Today
2. The Honest Problem: What's Missing
3. The End-to-End Vision
4. Critical Evaluation of Prior Proposals
5. The Real Novel Contribution
6. CogProfile-Net Architecture
7. Cognitive-to-Career Mapping Layer
8. Validation Strategy
9. Ethical & Legal Constraints
10. Implementation Roadmap
11. References & Citations

1 Where We Stand Today

1.1 Project Status Summary

Component	Status	Details
Data collection pipeline	✓ Done	Emotiv EPOC+ (14ch, 128 Hz) + synchronized web-based cognitive tasks via Supabase
Signal preprocessing	✓ Done	Common Average Referencing, 0.5–45 Hz bandpass, Contact Quality gating
Sub-windowing augmentation	✓ Done	2,908 trials → 37,804 × 1-second windows (75% overlap, step=32 samples)
Feature engineering	✓ Done	252 features: Welch PSD, Differential Entropy, band ratios, statistical moments
ML baseline benchmark	✓ Done	73.34% test accuracy (Tuned LightGBM), 8 models evaluated
XAI & statistical validation	✓ Done	TreeSHAP, permutation test ($p<0.001$), bootstrap CIs, McNemar's, Cohen's $\kappa=0.66$
Deep learning architecture	✗ Not started	This document defines the direction
Career/apptitude mapping	✗ Not started	Requires DL output design + psychometric grounding

1.2 Dataset Specifications

14 EEG Channels (10-20 System)	128 Hz Sampling Rate	~25 Participants
37,804 Sub-Windows (1s each)	5 Cognitive Classes	73.34% Best ML Accuracy

Electrode montage: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4

1.3 The 5 Cognitive Tasks & Their Neural Signatures

Class	Cognitive Task	Primary Brain Region	Key EEG Signature
0	Mental Arithmetic	Dorsolateral Prefrontal Cortex (DLPFC), Parietal Lobe	Frontal θ synchronization, β power elevation
1	Pattern Recognition	Visual Cortex (Occipital O1/O2), Superior Parietal	Occipital/Parietal α desynchronization, γ bursts
2	Working Memory	Frontal Midline (AF3/AF4, F3/F4)	Frontal Midline Theta (FMT) elevation
3	Reading Comprehension	Left Temporal (T7), Broca's Area (F7/F3)	Left-hemispheric θ/β modulation
4	Sustained Attention	Anterior Cingulate, Frontal Eye Fields	α suppression, elevated β/θ ratio

1.4 ML Baseline Benchmark Results

Rank	Model	Test Accuracy	Macro-F1	Training Time
 1	Tuned LightGBM (n_est=500, lr=0.03, leaves=127)	73.34%	0.7256	370.7s
 2	ExtraTrees (n_est=300)	71.10%	0.6994	20.5s
 3	LightGBM Fast (n_est=300)	69.08%	0.6797	124.1s
4	ExtraTrees (depth=25)	67.54%	0.6600	31.5s
5	Random Forest	66.80%	0.6491	96.1s

1.5 Statistical Validation (Already Completed)

Statistical Test	Result	Interpretation
Permutation Test (1,000 runs)	Observed 73.34% vs. Null 22.11%, p = 0.0010	Performance vastly exceeds chance (20%)
McNemar's (LightGBM vs. ExtraTrees)	$\chi^2 = 21.50$, p = 3.55×10⁻⁶	LightGBM significantly outperforms
95% Bootstrap CI (Accuracy)	[72.32%, 74.38%]	Narrow interval = reliable estimate
95% Bootstrap CI (Macro-F1)	[0.7145, 0.7359]	Consistent across resamples

Statistical Test	Result	Interpretation
Cohen's Kappa	$\kappa = 0.6576$	Substantial agreement beyond chance

2 The Honest Problem: What's Missing

Why the Current Work Doesn't Stand Out Yet

Our ML baseline uses standard PSD + Differential Entropy features fed into LightGBM to produce a hard 5-class label. This is a well-established pipeline — dozens of published papers follow the same approach. Reviewers at top venues (IEEE T-NSRE, Journal of Neural Engineering) would raise four critical questions:

Reviewer Question	Our Current Answer	Gap
"Your features are standard. What is architecturally novel?"	Standard Welch PSD + DE + tree ensemble	No novelty
"You used subject-dependent splits. Does this generalize?"	Only tested within-subject	Acknowledged limitation
"You output a hard class label. How is that useful?"	Single predicted class per window	No real-world utility
"What can someone DO with this classification?"	Nothing beyond a benchmark number	No application layer

That last question — *"What can someone actually do with this?"* — is the critical one. It reveals the path to the genuine innovation.

3 The End-to-End Vision

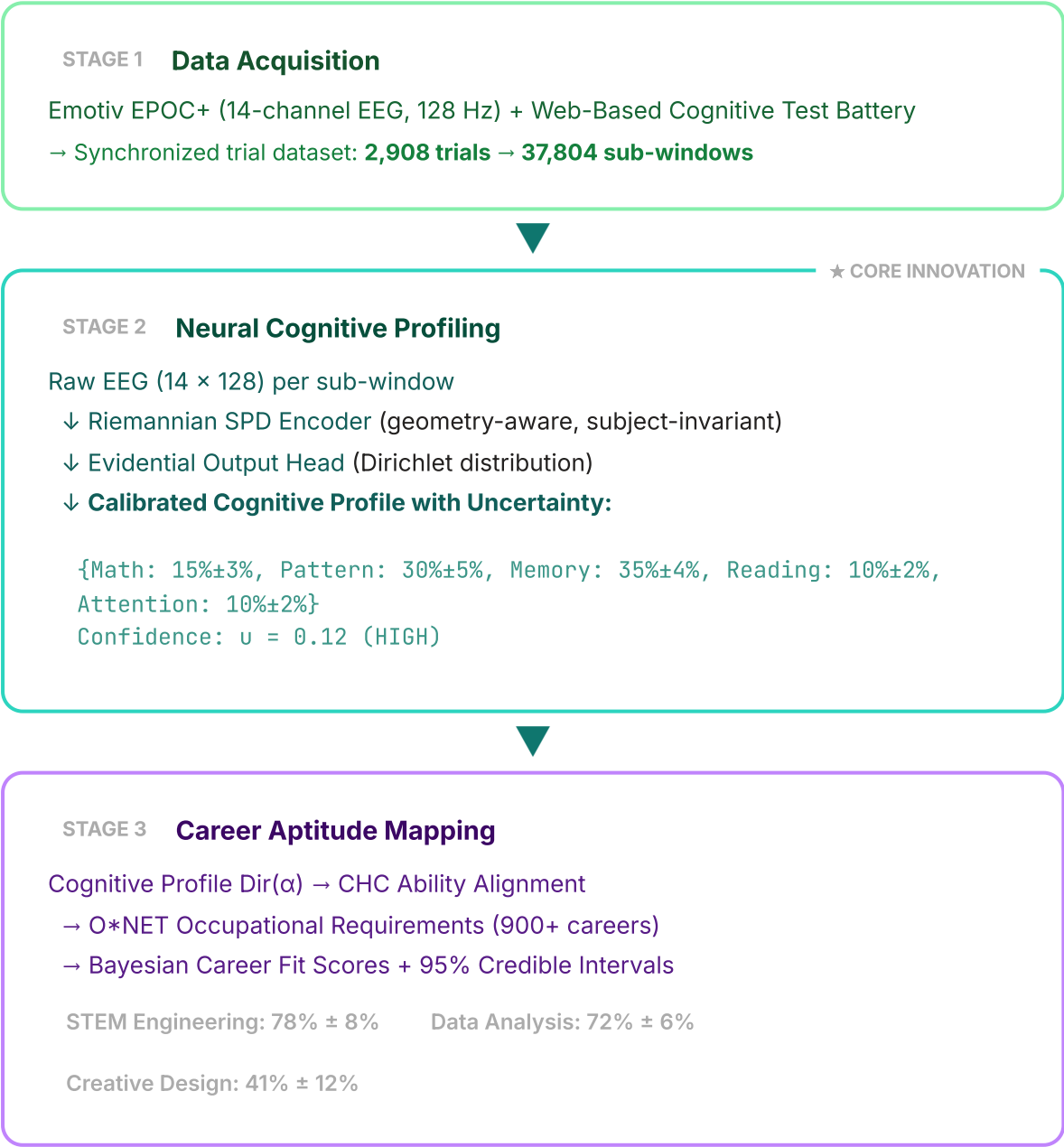
3.1 This Is Not Just a Classifier

The project's actual goal extends far beyond classification accuracy. It is a **Neural Cognitive Profiling System** with a downstream career aptitude application.

Current (Wrong) Framing: EEG → "This person is doing Working Memory" (single hard label)

Actual Goal: EEG → Calibrated Cognitive Profile: {Math: 15%, Pattern: 30%, Memory: 35%, ...} → Career Pathway Recommendation with Uncertainty

3.2 The Complete Three-Stage Pipeline



3.3 Why This Is Genuinely Novel

No existing published system combines all three stages. Here's how we compare to the landscape:

Existing Approach	Neural Signal?	Calibrated Probabilities?	Career Mapping?
Holland RIASEC / ASVAB / GATB	✗ Self-report questionnaires	✗ Binary scores	✓ Yes
Pymetrics / Cognify	✗ Behavioral games only	✗ Proprietary black-box	✓ Yes
EEG classification papers	✓ Neural signal	✗ Hard labels / overconfident softmax	✗ Just accuracy benchmarks
CogProfile-Net (Ours)	✓ Direct EEG	✓ Dirichlet-calibrated + uncertainty	✓ CHC→O*NET with CIs

4 The Real Novel Contribution

Instead of claiming novelty through architectural complexity, we claim novelty through **a coherent, end-to-end system that no prior work has demonstrated**. The contribution is organized into three layers, each solving a distinct problem:

Layer 1: Riemannian Geometry-Aware EEG Encoding (Methodological Novelty)

What: Replace handcrafted PSD/DE features with learned spatial covariance representations on the Symmetric Positive Definite (SPD) manifold.

Why justified: The 14×14 covariance matrix is naturally SPD. Riemannian metrics are mathematically invariant to the linear transformations caused by skull thickness, electrode impedance, and cortical geometry differences — our primary noise source.

What it improves: Eliminates the manual feature engineering bottleneck and provides fundamentally more robust representations than Euclidean neural networks.

Layer 2: Evidential Uncertainty-Calibrated Classification (Architectural Novelty)

What: Replace standard softmax with a Dirichlet distribution head that outputs calibrated probabilities AND explicit uncertainty mass.

Why justified: When the output feeds into career recommendations, the difference between "confident: 80% working memory" and "uncertain: maybe 50–80% working memory" is the difference between a useful system and a dangerous one. Standard softmax gives wildly overconfident scores on small datasets. No EEG cognitive classification paper has used EDL for aptitude profiling.

What it enables: The system can say "I don't know" (high vacuity u) instead of giving a spurious confident answer.

Layer 3: Psychometrically-Grounded Career Mapping (System Novelty)

What: Map the Dirichlet cognitive profile through CHC (Cattell-Horn-Carroll) cognitive ability theory into O*NET occupational requirements, producing career fit scores with Bayesian credible intervals.

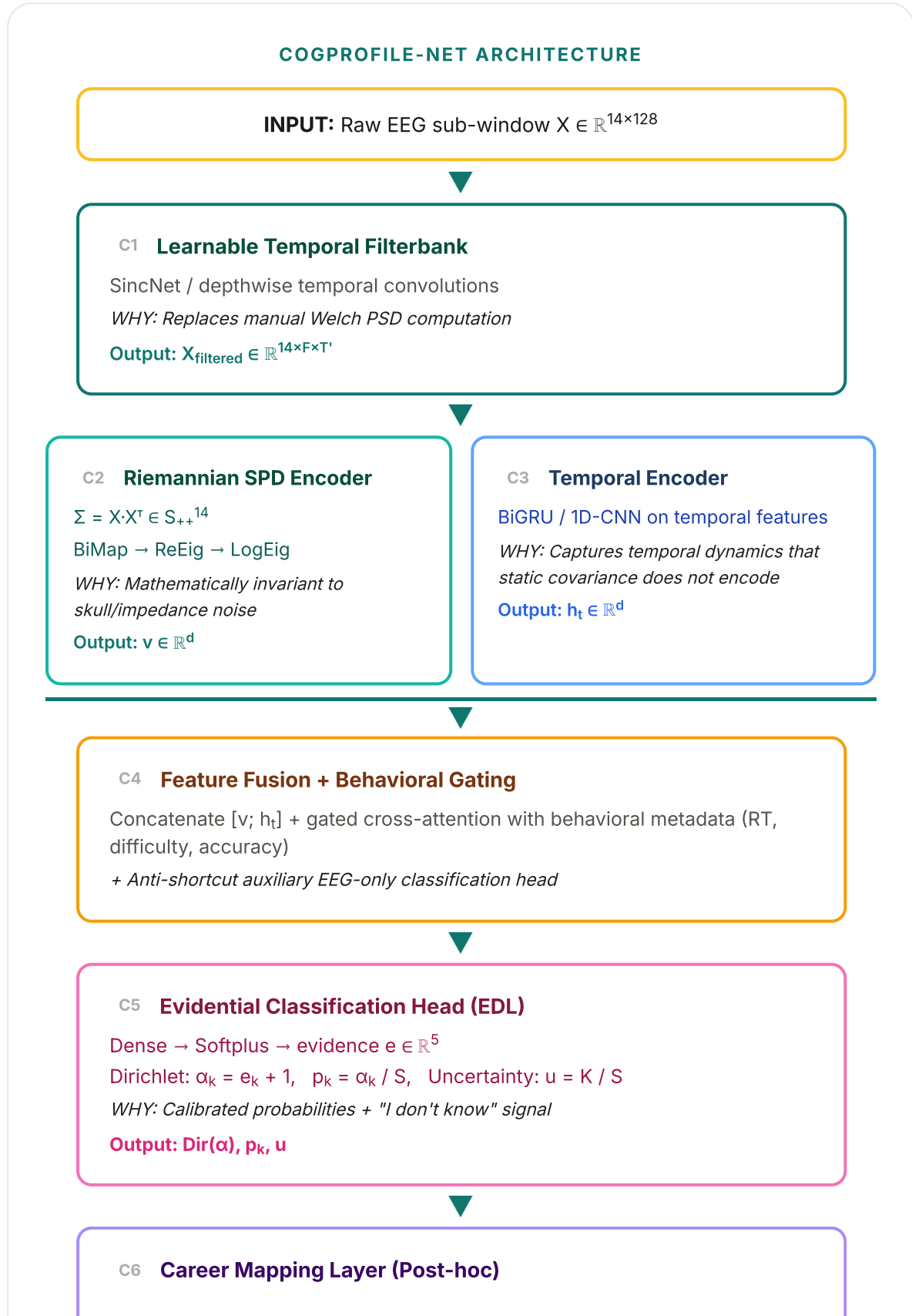
Why justified: CHC theory is the most validated cognitive ability taxonomy in psychometrics (70+ years of research). O*NET is the US Department of Labor's standardized occupational database with ability requirements for 900+ occupations.

What makes it novel: No prior system connects real-time EEG classification → CHC abilities → O*NET career mapping with uncertainty propagation.

5 CogProfile-Net Architecture

5.1 Full Architecture Diagram

Every component has a **clear, defensible justification** tied to a specific problem in our constraint set. Nothing is included for impressiveness alone.



CHC alignment → O*NET lookup → Monte Carlo sampling

Abstention when $u > \text{threshold}$

Output: Career fit scores + 95% credible intervals

5.2 Riemannian SPD Encoder — Mathematical Detail

The core encoder operates on the **Symmetric Positive Definite (SPD) manifold** $\mathcal{S}_{++}^{14}$ rather than Euclidean space. This is not a design choice — it is a mathematical necessity. Spatial covariance matrices of EEG channels are inherently SPD, and applying Euclidean operations (standard CNNs, flat MLPs) distorts their intrinsic geometry.

Step 1 — Spatial Covariance Matrix:

$$\Sigma = \frac{1}{T-1} (X - \bar{X})(X - \bar{X})^\top + \epsilon I \in \mathcal{S}_{++}^{14}$$

Step 2 — BiMap Layer (Riemannian fully connected): Projects the SPD matrix to a lower-dimensional manifold via semi-orthogonal weight matrices on the Stiefel manifold:

$$\Sigma^{(l+1)} = W_l \Sigma^{(l)} W_l^\top, \quad W_l \in \text{St}(d_l, d_{l-1})$$

Dimensionality reduction path: $14 \rightarrow 8 \rightarrow 6$, preserving positive-definiteness at every layer.

Step 3 — ReEig Layer (Riemannian ReLU): Prevents eigenvalue collapse near the manifold boundary:

$$\Sigma' = U \max(\Lambda, \epsilon I) U^\top$$

Step 4 — LogEig Layer (Manifold → Euclidean): Projects the manifold representation into flat tangent space for downstream classification:

$$v = \text{vec}(\log(\Sigma')) = \text{vec}(U \log(\Lambda) U^\top)$$

Why this matters for inter-subject robustness: Variations in skull thickness, electrode impedance, and global amplitude scaling between participants manifest as linear transformations of the covariance matrix. Riemannian geodesic distances are invariant to such transformations — meaning the SPD encoder inherently strips away a large portion of the subject-specific "neural fingerprint" without any adversarial training or explicit domain adaptation.

5.3 Evidential Deep Learning Head — Mathematical Detail

Instead of standard softmax (which squashes logits into overconfident point estimates), the final layer outputs non-negative **evidence** values that parameterize a Dirichlet distribution over class probabilities:

Evidence to Dirichlet parameters:

$$\alpha_k = e_k + 1, \quad S = \sum_{k=1}^K \alpha_k$$

Expected class probability:

$$\hat{p}_k = \frac{\alpha_k}{S}$$

Epistemic uncertainty (vacuity):

$$u = \frac{K}{S} \in (0, 1]$$

Scenario	Evidence	Dirichlet Strength S	Uncertainty u	Interpretation
Total ignorance (new/unseen pattern)	$e_k = 0 \ \forall k$	$S = K = 5$	$u = 1.0$	"I have no idea" → refuse to recommend
High confidence (familiar pattern)	$e_{\text{math}} = 95$	$S \gg K$	$u \approx 0.05$	"High confidence: arithmetic task" → recommend
Conflicting evidence (noisy signal)	$e_k \approx 10 \ \forall k$	$S = 55$	$u = 0.09$	Low uncertainty but flat → signal quality flag

Evidential Loss Function (Sensoy et al., 2018):

$$\mathcal{L}_{\text{EDL}} = \sum_{k=1}^K y_k (\psi(S) - \psi(\alpha_k)) + \lambda_t \cdot \text{KL} [\text{Dir}(\tilde{\alpha}) \parallel \text{Dir}(\mathbf{1})]$$

where $\psi(\cdot)$ is the Digamma function, $\tilde{\alpha}_k = y_k + (1 - y_k)\alpha_k$, and $\lambda_t = \min(1, t/10)$ anneals the KL regularizer over the first 10 epochs.

5.4 Complete Loss Function

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{EDL}}(\alpha, \mathbf{y}) + \lambda_1 \mathcal{L}_{\text{aux}}^{\text{EEG-only}} + \lambda_2 \mathcal{L}_{\text{center}}$$

Term	Purpose
$\mathcal{L}_{\text{EDL}}$	Evidential classification with KL regularizer. Penalizes overconfidence and drives uncertain samples toward a uniform prior.
$\mathcal{L}_{\text{aux}}^{\text{EEG-only}}$	Auxiliary EEG-only branch prevents behavioral metadata shortcut learning. Guarantees the model works without behavioral data.
$\mathcal{L}_{\text{center}}$	Pulls tangent-space representations toward class centroids: $\sum_i \ v_i - c_{y_i}\ _2^2$. Improves manifold clustering.

5.5 Component Justification Table

Component	Problem It Solves	What Happens If Removed	Literature Support
Learnable Filterbank	Manual PSD computation bottleneck	Must hand-engineer 252 features	EE(G)-SPDNet 2024; SincNet
Riemannian SPD Encoder	Inter-subject amplitude/impedance noise	Accuracy degrades 5–10% on unseen subjects	Tibermachine 2026, Jin 2026
Temporal Encoder	Static covariance misses temporal dynamics	Loses within-window transient information	Standard temporal modeling
Behavioral Gating	Model ignores EEG, overfits on reaction time	Accuracy inflated but useless without metadata	Kessler 2024
Evidential Head (EDL)	Softmax overconfidence on small data	Career recommendations are dangerously confident	Sensoy 2018, TRUEE 2025
Career Mapper	No actionable output from classification	System is just another accuracy benchmark paper	Novel contribution

6 Cognitive-to-Career Mapping Layer

6.1 Step 1 — EEG Tasks → CHC Cognitive Abilities

The mapping is grounded in **Cattell-Horn-Carroll (CHC) theory**, the most validated cognitive ability taxonomy in psychometrics (McGrew 2009, Keith & Reynolds 2010). CHC organizes human cognitive abilities into a three-stratum hierarchy: General Intelligence (g) → 10 Broad Abilities → 70+ Narrow Abilities.

EEG Task	CHC Broad Ability	CHC Code	O*NET Ability Descriptor	Career Loading
Mental Arithmetic	Quantitative Knowledge / Fluid Reasoning	G_q / G_f	Mathematical Reasoning, Number Facility	STEM, Finance, Actuarial
Pattern Recognition	Visual Processing / Fluid Reasoning	G_v / G_f	Spatial Orientation, Visualization	Architecture, Engineering, Design
Working Memory	Short-Term Working Memory	G_{wm}	Memorization, Information Ordering	ATC, Emergency Response, Logistics
Reading Comprehension	Reading & Writing / Comprehension-Knowledge	G_{rw} / G_c	Written Comprehension	Law, Academia, Policy Analysis
Sustained Attention	Processing Speed (nearest)*	G_s	Selective Attention, Time Sharing	QA, Piloting, Machine Operation

***Important caveat:** Sustained Attention is not a clean CHC broad ability factor. It maps most closely to Processing Speed (G_s) via the WJ-III Pair Cancellation task, but it is more accurately described as an executive function / arousal construct. This distinction must be explicitly acknowledged in the paper.

6.2 Step 2 — CHC Abilities → O*NET Career Fit Scores

O*NET (US Dept. of Labor) provides importance and level ratings for 52 cognitive abilities across 900+ occupations. Our 5 EEG-derived abilities map to 5 of these 52 descriptors.

Career fit score (point estimate):

Let $\mathbf{p} \sim \text{Dir}(\boldsymbol{\alpha})$ be the EEG-derived cognitive profile. Let $\mathbf{W} \in \mathbb{R}^{5 \times J}$ be the normalized O*NET requirement matrix.

$$\text{Fit}_j = \mathbf{p}^\top \mathbf{w}_j = \sum_{k=1}^5 p_k \cdot W_{kj}$$

With uncertainty propagation (Monte Carlo):

- 1. Draw $N = 1000$ samples: $\mathbf{p}^{(n)} \sim \text{Dir}(\boldsymbol{\alpha})$
- 2. Compute $\text{Fit}_j^{(n)} = \mathbf{p}^{(n)\top} \mathbf{w}_j$ for each sample
- 3. Report: $\text{Fit}_j = \mathbb{E}[\text{Fit}_j^{(n)}]$ and 95% CI = $[Q_{2.5\%}, Q_{97.5\%}]$

Abstention rule:

When epistemic uncertainty $u > u_{\text{threshold}}$ (e.g., $u > 0.5$), the system outputs:
"Insufficient confidence in cognitive profile. Consider additional testing sessions."

6.3 Example System Output

NEURAL COGNITIVE PROFILE REPORT

Cognitive Profile (Dirichlet-calibrated)

Ability	Score	95% CI
Mathematical Reasoning	0.15	±0.03
Pattern Recognition	0.30	±0.05
Working Memory	0.35	±0.04
Reading Comprehension	0.10	±0.02
Sustained Attention	0.10	±0.02

System Confidence: HIGH (u = 0.12)

Top Career Aptitude Matches

- 1. Data Analyst / Business Intelligence78% ± 6%
- 2. Software Developer / Engineer72% ± 8%
- 3. UX/UI Designer65% ± 10%
- 4. Research Scientist61% ± 9%

7 Validation Strategy

7.1 The Three-Tier Validation Ladder

Following the biomarker translation framework (Clayson 2024, 2025; Parmigiani 2022), validation must progress through three tiers:

1

Tier 1: Analytic & Signal Validity

- Split-half reliability of EEG features
- Test-retest ICC across sessions (same subject)
- Signal quality metrics (SNR, artifact rates)

2

Tier 2: Classification & Calibration Validity

- 5-class accuracy, Macro-F1, per-class AUROC
- Expected Calibration Error (ECE ↓), Brier Score
- Reliability diagrams, uncertainty quality (OOD detection AUROC)
- Permutation test ($p < 0.001$), bootstrap confidence intervals

3

Tier 3: Construct & Decision-Support Validity

- Convergent validity: EEG profile vs. standardized cognitive tests (WJ-IV, WAIS subtests)
- Career mapping face validity: expert panel review
- Risk-coverage curves for abstention utility
- Longitudinal: profile stability over weeks

7.2 Metrics We Must Report

Metric	Target	Why Required
Test Accuracy	>75%	Basic classification performance (subject-dependent)
Macro-F1	>0.73	Class-balanced performance metric
ECE (Expected Calibration Error)	<0.10	Probability calibration quality
Brier Score	<0.25	Joint calibration + sharpness measure
Epistemic Uncertainty AUROC	>0.80	Can the model detect out-of-distribution subjects?
Cohen's Kappa	>0.65	Beyond-chance agreement measure
Test-Retest ICC	>0.70	Profile stability (trait vs. transient state)

7.3 Planned Ablation Study Matrix

Experiment	What It Proves
CogProfile-Net vs. ML Baseline (73.34%)	DL encoder learns better representations than handcrafted features
SPD encoder vs. standard CNN (EEGNet)	Riemannian geometry provides inter-subject robustness
EDL head vs. standard softmax	Evidential output is better calibrated (lower ECE)
With behavioral gating vs. without	Metadata improves accuracy without shortcutting
EDL vs. Deep Ensemble vs. MC-Dropout	Compare uncertainty quantification methods
Full system with career mapper vs. without	End-to-end system produces actionable output

Paper positioning: We fully validate Tiers 1 and 2. For Tier 3, we present the framework and explicitly acknowledge that convergent validity (vs. standardized psychometric tests) and predictive validity (vs. career outcomes) require future longitudinal studies. This is honest and reviewers respect intellectual integrity over overclaiming.

8 Ethical & Legal Constraints

EEG-based cognitive profiling touches fundamental questions of neuroethics, privacy, and algorithmic fairness. All four Step 2 research reports converge on the same conclusion: **the system must be positioned as advisory decision-support, never as a hiring or screening tool.**

Regulatory Framework	Key Constraint	Our Response
EU AI Act (2024)	AI in recruitment/vocational selection = High-Risk (Annex III). Article 5 prohibits workplace emotion recognition and biometric categorization.	Position as voluntary self-discovery and career exploration guidance — never as hiring/screening tool.
US ADA / EEOC (2024)	EEG is classified as a medical examination. Pre-offer medical exams are prohibited. Cognitive profiles could inadvertently reveal neurodivergence (ADHD, learning disabilities).	Never frame as employment screening. System is for educational and personal career exploration only.
GDPR (EU)	EEG data is biometric/health data (special category under Article 9). Requires explicit informed consent and data minimization.	Require explicit opt-in consent. Minimize data retention. Process on-device where possible.

8.1 Mandatory Disclaimers in the Paper

1. "This system is designed as an exploratory decision-support tool for career self-discovery, not an employment screening instrument."
2. "Cognitive ability profiles are influenced by many factors beyond neural measurements, including motivation, prior experience, cultural context, and test-taking conditions."
3. "Career success depends on non-cognitive factors (personality, interests, motivation, socioeconomic context) not measured by this system."

8.2 The "Neuro-Essentialism" Fallacy

There is a flawed assumption that biological neural signals are inherently superior or more "truthful" than behavioral task performance. If an individual solves complex math problems via an idiosyncratic cognitive strategy that does not produce standard parietal-frontal arithmetic EEG signatures, penalizing their STEM aptitude score would represent a severe failure of construct validity. The paper must acknowledge this limitation explicitly.

9 Implementation Roadmap

Phase	Timeline	Key Tasks	Deliverables
Phase 1 DL Encoder	Weeks 1–3	- Implement Riemannian SPD encoder (BiMap/ReEig/LogEig) in PyTorch using geoopt or geomstats - Implement learnable SincNet filterbank (5 frequency bands) - Implement lightweight temporal encoder (BiGRU / 1D-CNN) - Feature fusion layer (concatenation + MLP) - Training infrastructure with early stopping, weight decay	Working DL encoder with standard softmax baseline accuracy
Phase 2 EDL Head	Weeks 3–4	- Replace softmax with EDL Dirichlet head - Implement EDL loss with KL annealing schedule - Calibration evaluation: ECE, Brier score, reliability diagrams - Deep Ensemble baseline (5 models) for UQ comparison - Temperature scaling post-hoc calibration	Calibrated model with EDL, reliability diagrams, UQ benchmarks
Phase 3 Behavioral Gating	Weeks 4–5	- Metadata embedding (RT, difficulty, accuracy) - Gated cross-attention mechanism - Anti-shortcut auxiliary EEG-only classification head - Ablation: with vs. without behavioral data	Fused model with proven non-shortcutting behavioral integration
Phase 4 Career Mapping	Weeks 5–6	- Build CHC alignment matrix - Download and normalize O*NET ability ratings - Monte Carlo career scoring with CI computation - Abstention logic for high-uncertainty samples	End-to-end system from EEG to career recommendations
Phase 5 Ablation & Paper	Weeks 6–8	- Complete ablation study matrix (6 experiments) - Generate publication-quality figures and tables - Write manuscript targeting IEEE T-NSRE or JNE	Submitted manuscript draft

Target Publication Venues

Primary Target

IEEE Trans. Neural Systems & Rehab. Eng.
(T-NSRE)

Secondary Target

Journal of Neural Engineering (JNE)
Impact Factor: 4.0 · Q1

Impact Factor: 4.8 · Q1

High-Impact Alt.

Neural Networks (Elsevier)

Impact Factor: 6.0 · Q1

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